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Unit 2 Assignment: Building a Mixture of Experts (MoE) Router using Groq API

The goal of this assignment was to design and implement a **Smart Customer Support Router** using a **Mixture of Experts (MoE)** architecture. Instead of using a single general-purpose AI model, we created specialized "experts" for different types of user queries such as technical issues, billing concerns, and general conversations.

This architecture mimics real-world enterprise AI systems where domain-specific handling improves accuracy and user experience.

The system consists of three main components:

1. Router (Intent Classifier)
2. Experts (Specialized System Prompts)
3. Orchestrator (Controller Logic)

Flow of execution:

1) User Input
2) Router (LLM with temperature = 0)
3) Category Selection
4) Expert Model (LLM with temperature = 0.7)
5) Final Response

Expert Configuration

We defined a configuration dictionary (MODEL_CONFIG) to simulate multiple experts using different system prompts, while keeping the same base model.

Experts Implemented:

- Technical Expert - Rigorous, code-focused, step-by-step debugging guidance.
- Billing Expert : Empathetic tone, policy-driven, refund/subscription handling.
- General Expert: Friendly fallback assistant for casual conversations.

Router Design:

The router uses an LLM call to classify user input into one of: [technical, billing, general]

- **Low temperature (0)** : very predictable, safest token choice
- **Medium temperature (0.7)** : balanced creativity and coherence
- **High temperature (>1)** : highly random and unpredictable

temperature = 0, Ensures deterministic classification.

Orchestrator Logic

The process_request() function performs:

1. Calls route_prompt() to classify the input.
2. Loads the appropriate system prompt.
3. Calls the LLM with temperature = 0.7 for flexible response generation.
4. Returns the expert's answer.

This separation ensures modularity and scalability.

Bonus: Tool Use Expert

We implemented a simple tool-based routing mechanism:

If the query contains: "bitcoin" AND "price"

The system bypasses the LLM and calls a mock function to fetch data.

This demonstrates how MoE systems can integrate:

- External APIs
- Real-time data
- Function calling mechanisms

The Architecture is Modular and scalable design, Specialized expertise per domain and Production-ready structure

Outcomes:

Using temperature strategically for different tasks, Implementing LLM-based intent classification. Designing modular AI systems and Integrating tool usage into LLM pipelines.

Conclusion

In this assignment, we successfully built a working Mixture of Experts Router using the Groq API. The system dynamically routes user queries to specialized expert configurations, demonstrating how advanced LLM architectures are implemented in real-world AI systems.

This project highlights the importance of structured AI orchestration over using a single general-purpose model.